

Yıldız Technical University
Computer Engineering Department
2023-2024 Spring
BLM3620 Digital Signal Processing
Homework 4, Form: A

Name: _____

Surname: _____

Student I.D.: _____

Signature: _____

Signal	FT
$x(t)$	$X(w)$
$y(t)$	$Y(w)$
$\frac{1}{2\pi} \int_{-\infty}^{\infty} X(w)e^{jw t} dw$	$\int_{-\infty}^{\infty} x(t)e^{-jw t} dt$
$\delta(t)$	1
$\Pi(t) = \begin{cases} 0, & t > \frac{1}{2} \\ 1, & t \leq \frac{1}{2} \end{cases}$	$\text{sinc}\left(\frac{w}{2\pi}\right)$
$\Lambda(t) = \begin{cases} 0, & t > 1 \\ 1 - t , & t \leq 1 \end{cases}$	$\text{sinc}^2\left(\frac{w}{2\pi}\right)$
1	$2\pi\delta(w)$
$e^{-jw_0 t}$	$2\pi\delta(w - w_0)$
$e^{- a t}u(t)$	$\frac{1}{ a + jw}$
$e^{ a t}u(-t)$	$\frac{1}{ a - jw}$
$e^{- at }$	$\frac{2 a }{ a ^2 + w^2}$
$e^{-\pi t^2}$	$e^{-\frac{w^2}{4\pi}}$
$u(t)$	$\pi\delta(w) + \frac{1}{jw}$
$\cos(w_0 t)$	$\pi\delta(w + w_0) + \pi\delta(w - w_0)$
$\sin(w_0 t)$	$j\pi\delta(w + w_0) - j\pi\delta(w - w_0)$
$\sum_{n=-\infty}^{\infty} \delta(t - n)$	$\sum_{n=-\infty}^{\infty} \delta(w - n)$
$t^n x(t)$	$j^n \frac{d^n X(w)}{dw^n}$
$ t $	$-\frac{1}{w^2}$
$\frac{1}{1+t^2}$	$\pi e^{- w }$
$ax(t) + by(t)$	$aX(w) + bY(w)$
$x(t - a)$	$X(w)e^{-jwa}$
$x\left(\frac{t}{a}\right)$	$ a X(aw)$
$x(t) * y(t)$	$X(w)Y(w)$
$x(t)y(t)$	$\frac{1}{2\pi} X(w) * Y(w)$
$x(t)e^{jw_0 t}$	$X(w - w_0)$
$x^*(t)$	$X^*(-w)$
$x(-t)$	$X(-w)$
$\frac{dx(t)}{dt}$	$jwX(w)$
$\int_{-\infty}^t x(\tau) d\tau$	$\frac{1}{jw} X(w)$
$x(t)$ is real	$X(w) = X^*(-w)$
$x(t)$ is real and even	$X(w)$ is real and even
$x(t)$ is real and odd	$X(w)$ is imaginary and odd
$\int_{-\infty}^{\infty} x(t) ^2 dt$	$\frac{1}{2\pi} \int_{-\infty}^{\infty} X(w) ^2 dw$

Sequence	DTFT
$x[n]$	$X(e^{jw})$
$y[n]$	$Y(e^{jw})$
$\frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{jw}) e^{jwn} dw$	$\sum_{n=-\infty}^{\infty} x[n] e^{-jwn}$
$\delta[n]$	1
$\delta[n-a]$	e^{-jwa}
$\sum_{m=-\infty}^{\infty} \delta[n-m]$	$2\pi \sum_{k=-\infty}^{\infty} \delta(w-2\pi k)$
e^{-jan}	$2\pi \sum_{k=-\infty}^{\infty} \delta(w+a-2\pi k)$
$u[n]$	$\frac{1}{1-e^{-jw}} + \pi \sum_{k=-\infty}^{\infty} \delta(w-2\pi k)$
$a^n u[n], a < 1$	$\frac{1}{1-ae^{-jw}}$
$-a^n u[-n-1], a > 1$	$\frac{1}{1-ae^{-jw}}$
$(n+1)a^n u[n], a < 1$	$\frac{1}{(1-ae^{-jw})^2}$
$\cos(na)$	$\pi \sum_{k=-\infty}^{\infty} \delta(w-a-2\pi k) + \delta(w+a-2\pi k)$
$\sin(na)$	$\frac{\pi}{j} \sum_{k=-\infty}^{\infty} \delta(w-a-2\pi k) - \delta(w+a-2\pi k)$
$x[n] = \begin{cases} 0, & n > N \\ 1, & n \leq N \end{cases}$	$\frac{\sin(w(N+\frac{1}{2}))}{\sin(\frac{w}{2})}$
$\frac{\sin(An)}{\pi n}$	$X(e^{jw}) = \begin{cases} 1, & w \leq A \\ 0, & A < w \leq \pi \end{cases}, X(e^{jw}) \text{ periodic by } 2\pi$
$ax[n] + by[n]$	$aX(e^{jw}) + bY(e^{jw})$
$x[n-n_0]$	$X(e^{jw}) e^{-jwn_0}$
$x[n] e^{jwn_0}$	$X(e^{j(w-w_0)})$
$x^*[n]$	$X^*(e^{-jw})$
$x[-n]$	$X(e^{-jw})$
$x[n] * y[n]$	$X(e^{jw}) Y(e^{jw})$
$x[n] y[n]$	$\frac{1}{2\pi} \int_{2\pi} X(e^{j\theta}) Y(e^{j(w-\theta)}) d\theta$
$x[n] - x[n-1]$	$(1-e^{-jw}) X(e^{jw})$
$\sum_{k=-\infty}^{\infty} x[k]$	$\frac{1}{(1-e^{-jw})} + \pi X(1) \sum_{k=-\infty}^{\infty} \delta(w-2\pi k)$
$nx[n]$	$j \frac{dX(e^{jw})}{dw}$
$x[n]$ is real	$X(e^{jw}) = X^*(e^{-jw})$
$x[n]$ is real and even	$X(e^{jw})$ is real and even
$x[n]$ is real and odd	$X(e^{jw})$ is imaginary and odd
$\sum_{n=-\infty}^{\infty} x[n] ^2$	$\frac{1}{2\pi} \int_{-\infty}^{\infty} X(e^{jw}) ^2 dw$

Sequence	DFT
$x[n]$	$X[k]$
$y[n]$	$Y[k]$
$\frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j2\pi \frac{kn}{N}}$	$\sum_{n=0}^{N-1} x[n] e^{-j2\pi \frac{kn}{N}}$
$\sum_{k=-\infty}^{\infty} \delta[n + Nk]$	1 (period N)
1 (period N)	$N \sum_{m=-\infty}^{\infty} \delta[k + Nm]$
$e^{j2\pi k_0 n}$	$N \delta[((k - k_0))_N]$
$\cos\left(2\pi \frac{k_0 n}{N}\right)$	$\frac{N}{2} (\delta[((k - k_0))_N] + \delta[((k + k_0))_N])$
$ax[n] + by[n]$	$aX[k] + bY[k]$
$x[((n - m))_N]$	$X[k] e^{-j2\pi \frac{km}{N}}$
$X[n]$	$NX[((-k))_N]$
$x[n]y[n]$	$\frac{1}{N} X[k] \otimes Y[k]$
$x[n] \otimes y[n]$	$X[k]Y[k]$
$x^*[n]$	$X^*[((-k))_N]$
$x[((-n))_N]$	$X^*[k]$
$\text{Re}\{x[n]\}$	$\frac{1}{2} (X[((k))_N] + X^*[((-k))_N])$
$j \text{Im}\{x[n]\}$	$\frac{j}{2} (X[((k))_N] - X^*[((-k))_N])$
$\frac{1}{2} (x[((n))_N] + x^*[((-n))_N])$	$\text{Re}\{X[k]\}$
$\frac{j}{2} (x[((n))_N] - x^*[((-n))_N])$	$j \text{Im}\{X[k]\}$
$\sum_{n=0}^{N-1} x[n] ^2$	$\frac{1}{N} \sum_{k=0}^{N-1} X[k] ^2$

Sequence	ZT	Region of Convergence
$x[n]$	$X(z)$	
$y[n]$	$Y(z)$	
$a^n u[n]$	$\frac{1}{1 - az^{-1}}$	$ z > a $
$-a^n u[-n - 1]$	$\frac{1}{1 - az^{-1}}$	$ z < a $
$na^n u[n]$	$\frac{az^{-1}}{(1 - az^{-1})^2}$	$ z > a $
$-na^n u[-n - 1]$	$\frac{az^{-1}}{(1 - az^{-1})^2}$	$ z < a $
$\delta[n]$	1	All z
$\delta[n - n_0]$	z^{-n_0}	All z
$u[n]$	$\frac{1}{1 - z^{-1}}$	$ z > 1$
$\cos(\omega_0 n) u[n]$	$\frac{1 - z^{-1} \cos(\omega_0)}{1 - 2z^{-1} \cos(\omega_0) + z^{-2}}$	$ z > 1$
$\sin(\omega_0 n) u[n]$	$\frac{z^{-1} \sin(\omega_0)}{1 - 2z^{-1} \cos(\omega_0) + z^{-2}}$	$ z > 1$
$a^n \cos(\omega_0 n) u[n]$	$\frac{1 - az^{-1} \cos(\omega_0)}{1 - 2az^{-1} \cos(\omega_0) + a^2 z^{-2}}$	$ z > a $
$a^n \sin(\omega_0 n) u[n]$	$\frac{az^{-1} \sin(\omega_0)}{1 - 2az^{-1} \cos(\omega_0) + a^2 z^{-2}}$	$ z > a $
$Ax[n] + By[n]$	$AX(z) + BY(z)$	
$x[n - n_0]$	$X(z)z^{-n_0}$	
$a^n x[n]$	$X(a^{-1}z)$	
$x^*[n]$	$X^*(z^*)$	
$x[-n]$	$X(z^{-1})$	
$x[n] * y[n]$	$X(z)Y(z)$	
$nx[n]$	$-z \frac{dX(z)}{dz}$	
$x[n]$ is causal	$x(0) = \lim_{z \rightarrow \infty} X(z)$	
$x[n]$ is causal	$x(\infty) = \lim_{z \rightarrow 1} [z - 1]X(z)$	

1. Determine the fundamental period of $x[n]$.

$$x[n] = \cos\left(\frac{n\pi}{10}\right) + \sin\left(\frac{n\pi}{15}\right)$$

2. Determine whether or not the signal $x[n]$ is periodic.

$$x[n] = \sin(\sqrt{2} + 0.2n)$$

3. Given that real valued signal $x_1[n]$ is even by definition $x_1[n] = x_1[-n]$, and real valued signal $x_2[n]$ is odd by definition $x_2[n] = -x_2[-n]$, determine symmetry (even/odd) of $y[n]$.

$$y[n] = x_1[n] \cdot x_2[n]$$

4. Given that the power of real valued signal $x[n]$ is defined as $P = \sum_{n=-\infty}^{\infty} x^2[n]$, compute the power in $y[n]$.

$$y[n] = 2^n \cdot u[-n]$$

5. Given that $x[n]$ is the system input and $y[n]$ is the system output, determine whether or not the following systems is time (shift)-invariant.

$$y[n] = x[n] \cdot u[n]$$

6. Given that $x[n]$ is the system input and $y[n]$ is the system output, determine whether or not the following systems is linear.

$$y[n] = \text{Im}(x[n])$$

7. Given that $x[n]$ is the system input and $y[n]$ is the system output, determine whether or not the following systems is casual.

$$y[n] = x[|n|]$$

8. Given that $x[n]$ is the system input and $y[n]$ is the system output, determine unit sample response $h[n]$ of the system.

$$y[n] = 0.5y[n-1] + 4x[n-2]$$

9. The responses of a linear time (shift)-invariant system to specified inputs are defined as follows:

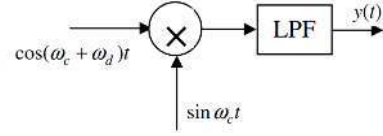
input		reponse
name	symbol	
unit sample	$\delta[n]$	$h[n]$
unit step	$u[n]$	$s[n]$

calculate $h[n]$ for the system, given that $s[n] = u[n] - u[n-5]$.

10. Find the Fourier transform of $x(t)$.

$$x(t) = \begin{cases} \frac{1}{2}, & -T < t < T \\ 0, & \text{other} \end{cases}$$

11. Given that the cut-off frequency of the low pass filter (LPF) is w_c evaluate the output $y(t)$. Note: LPF allows frequency values between $-w_c < w < w_c$.



$$x[n] = 2\delta[n+2] - \delta[n+1] + 3\delta[n] - \delta[n-1] + 2\delta[n-2]$$

12. Given that $x[n] \xleftrightarrow{DTFT} X(e^{jw})$ is a DTFT pair, evaluate $X(e^{jw})|_{w=\pi}$ without explicitly finding $X(e^{jw})$.

$$x[n] = 2\delta[n+2] - \delta[n+1] + 3\delta[n] - \delta[n-1] + 2\delta[n-2]$$

13. Given that $x[n] \xleftrightarrow{DTFT} X(e^{jw})$ and $y[n] \xleftrightarrow{DTFT} Y(e^{jw})$ are DTFT pairs, prove the convolution theorem.

$$x[n] * y[n] \xleftrightarrow{DTFT} X(e^{jw})Y(e^{jw})$$

14. Find the inverse DTFT of $X(e^{jw})$.

$$X(e^{jw}) = \cos^2(w)$$

15. Given the 6-point sequence $x[n] = [4, -1, 4, -1, 4, -1]$, determine its 6-point DFT sequence $X[k]$.

16. If the 4-point DFT an unknown length-4 sequence $v[n]$ is $V[k] = [1, 4 + j, -1, 4 - j]$, determine $v[n]$.

17. Find z-transforms of $x[n]$.

$$x[n] = 6\delta[n] - 7\delta[n-3] - 2\delta[n] - 9\delta[n-5]$$

18. If the region of convergence (ROC) for any $x[n] \xleftrightarrow{ZT} X(z)$ z-transform pair includes the unit circle in the complex plane then, $x[n] \xleftrightarrow{DTFT} X(e^{jw})$ DTFT pair can also be calculated (converges). Given that the following $X(z)$ includes the unit circle in its region of convergence, evaluate DTFT of $x[n]$ at $w = \pi$.

$$X(z) = \frac{z + 2z^{-2} + z^{-3}}{1 - 3z^{-4} + z^{-5}}$$

19. Evaluate $h[n] * x[n]$ using the convolution property of z-transform.

$$h[n] = (0.5)^n u[n]$$

$$x[n] = 3^n u[-n]$$

20. Given that $x[n] \xleftrightarrow{ZT} X(z)$ is a z-transform pair find $x[n]$.

$$X(z) = 2 + 5(z^2 + z^{-2})$$

21. Given that $x[n] \xleftrightarrow{ZT} X(z)$ is a z-transform pair find $x[n]$ for $|z| > 2$.

$$X(z) = \frac{1}{1 + 3z^{-1} + 2z^{-2}}$$

22. A continuous-time sinusoid $a_1(t) = \cos(w_1 t + 0.1\pi)$ is sampled at $f_{s_1} = 40 \text{ Hz}$ to give $a_1[n]$, and a second continuous-time sinusoid $a_2(t) = \cos(w_2 t + 0.1\pi)$ is sampled at $f_{s_2} = 50 \text{ Hz}$ to give $a_2[n]$. If $w_2 = 30\pi \text{ rad/s}$, determine w_1 so that $a_1[n] = a_2[n]$. Assume there is no aliasing when sampling $a_1(t)$.

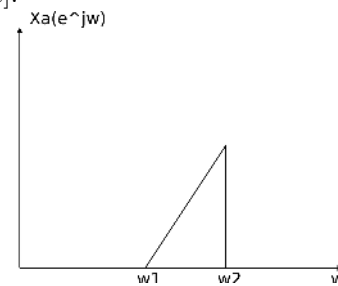
23. A complex bandpass filter is constructed by frequency shifting a running sum filter. evaluate and plot the magnitude frequency response of the complex bandpass filter $|H_B(e^{jw})|$.

$$h[n] = \sum_{k=0}^4 \delta[n - k]$$

$$h_B[n] = h[n]e^{jw_0 n}$$

$$h_B[n] \xleftrightarrow{DTFT} H_B(e^{jw})$$

24. A complex bandpass analog signal $x_a(t)$ has Fourier transform that is non-zero over the range of $[w_1, w_2]$. The signal is sampled to produce the sequence $x[n] = x_a(nT_s)$. What is the smallest sampling frequency that can be used so that $x_a(t)$ may be recovered from its samples $x[n]$.



25. Plot STFT (Short Time Fourier Transform) representation of a 1D chirp signal with different window sizes. Compare the results with FT of the same signal. Comment on what might be an optimal window size.